

TNPSC Mentor - Developer Handover

A bilingual (English/Tamil) TNPSC exam-prep web app: a large, subject-tagged question bank, auto-graded mock tests, AI-written explanations with per-option "why-wrong" feedback, analytics, spaced-repetition revision, and a self-updating current-affairs feed.

Question bank (live): ~12,700 questions, **94% bilingual**. A duplicate-text cleanup pass removes the old per-group PYQ triplication (~2,400 redundant rows), settling the unique total to ~10,300. Every question has a real explanation and a per-option `why_wrong` map.

Category	Code	~Count
Previous Year / subject-wise GK	<code>pyq</code>	8,556
Current Affairs	<code>current_affairs</code>	1,875
Samacheer (state board)	<code>samacheer</code>	1,189
Aptitude & Reasoning	<code>aptitude</code>	1,075

1. Tech stack

Layer	Technology
Frontend	Vite 5 · React 18 · TypeScript 5 · Tailwind CSS 3
Routing / state	React Router v6 · Zustand (auth, quiz, language)
Backend	Supabase - Postgres + Auth + RLS via PostgREST
PDF / export	jsPDF + html2canvas (in-app); PyMuPDF (doc generation)
Fonts	Rajdhani (headings), Inter (body), Noto Sans Tamil
Data pipeline	Python 3.11 · requests · BeautifulSoup/lxml · PyMuPDF · deep-translator · anthropic
AI	Claude (Opus 4.8 + Haiku 4.5) via the Messages Batches API
Automation	GitHub Actions (monthly current-affairs cron)

2. Question sources (by `source_url` domain)

All questions are MCQs with a verified answer, normalised into one schema. Each row stores the exact `source_url` it came from.

Source	~Count	Content	Bilingual	Scraper
IndiaBix	6,435	GK practice, aptitude, reasoning	partial (MT)	<code>pyq_scraper.py</code> , <code>aptitude_scraper.py</code> , <code>samacheer_scraper.py</code> , <code>indiabix_common.py</code>
TNPSC Master	2,662	Subject/topic-wise TNPSC Q&A with explanations	native	<code>tnpscmaster_scraper.py</code>
GKToday	1,504	Current affairs (month + topic)	MT	<code>current_affairs_scraper.py</code>

TNPSC Guru	1,207	Subject-wise TNPSC MCQs	native	tnpscguru_scraper.py
Shankar IAS / Thervu Pettagam	371	Monthly CA quiz PDFs	native	thervupettagam_scraper.py
PYQ-paper aggregators (tnpscjob, jagranjosh, adda247, collegedunia, drive)	~516	Genuine previous-year papers	partial	pyq_scraper.py

Plus a Hugging Face open dataset (snegha24/Tamil_tnpscExam) seeded native Tamil (tamil_dataset_loader.py). **Not used:** Sura Books (paid/login-gated) and Winmeen's Samacheer 6-12 (paid + TLS-blocked) - only Winmeen's free aptitude is reachable.

Scraper notes (non-obvious)

- **TNPSC Master** is a Next.js app - question data is in RSC stream chunks, not the DOM. Fetch each topic URL with header `RSC: 1`, reconstruct the stream, locate the `{"...", "questions": [...]}` object, resolve lazy `$NN` text refs.
- **GKToday** paginates with `?pageno=2..N` (10 MCQs/page) - walk until a page returns no new questions (don't stop at page 1).
- **TNPSC Guru** is Blogger - paginate the JSON feed by the *actual* entries returned (Blogger size-truncates pages), not a fixed step.
- **Thervu Pettagam** quiz docs are digital text PDFs (PyMuPDF); answer key is a `N. x` table at the end; EN then TA interleaved per question.

3. AI pipelines (anti-hallucination by design)

The model is always **anchored on the verified answer** - it explains/rewrites, it never decides. All use the Batches API (50% cheaper) + structured JSON output. Default model `claude-opus-4-8`; `EXPLANATION_MODEL` env overrides (Haiku used for bulk).

- **generate_explanations.py** - fills missing/placeholder explanations. Output: why the correct option is right + **"Why not the others: (A)...; (B)..."**.
- **build_why_wrong.py** - populates `questions.why_wrong` (jsonb) = `{"<wrong letter>": "why it's wrong"}`. Phase 1 parses the explanations above (free); Phase 2 generates the rest.
- **paraphrase_questions.py** - rewrites third-party stems + options into original wording (copyright de-risk), preserving every number/name and the answer. Mixed model (Opus for aptitude, Haiku else). Numeric-guard rejects any rewrite that changes a number; originals backed up to `paraphrase_backup.jsonl` (gitignored). Genuine previous-year-paper items are kept verbatim.
- **translate_to_tamil.py** - fills `*_ta` columns via free Google MT (`--mode questions / --mode explanations`). Native-Tamil source rows skip MT.
- **consolidate.py** - runs `paraphrase(new) -> why_wrong -> explanation-Tamil ->` dedupe sequentially (avoids concurrent-paraphrase / MT clashes).
- **dedupe_questions.py** - collapses duplicate `question_text`, keeping the richest copy (scores explanation/why_wrong/Tamil presence). `--apply` to delete.

4. Data model (supabase/schema.sql)

- **questions** - category, subject, topic, group/standard/CA/aptitude metadata, question_text, option_a..d, correct_answer, explanation, **why_wrong** (jsonb), source_url, and Tamil mirrors *_ta.
- **profiles** (role user/admin, exam_date, daily_goal), **test_sessions** / **test_answers**, **review_items** (SRS), **daily_activity** (streaks).
- **user_percentile** (uuid) SECURITY DEFINER RPC. RLS on all user tables.

5. Syllabus taxonomy (`src/lib/constants.ts`)

GS subjects extended to the full TNPSC syllabus (added **Geography, General Science**) plus language papers (**General Tamil, General English**) for

Group 2/4. **PYQ uses subject-membership** - a question is shown under any group whose syllabus includes its subject (`fetchQuestions.ts` filters by subject, not a stored `group_type`), so each question is stored once.

6. App structure (`src/`)

- **pages/** - LanguageScreen -> TestArena -> category pickers -> SetupPage -> QuizPage -> ResultPage; plus MockTest, Daily (CA), Revision (SRS), Insights, AdminQuestions (admins view the full list **with answers**), auth pages.
- **store/** - authStore, quizStore (timer, 80% gate, flagging, scoring), languageStore. **lib/** - fetchQuestions, analytics, srs, habit, assets, i18n, pdfGenerator, constants, supabase.
- **Quiz rules:** 45s/question, 15s min, 80%-attendance gate to unlock explanations + PDF, auto-submit. **Answers/explanations are never shown mid-quiz** - only consolidated on the Result page (correct answer + targeted "Why your answer (X) is wrong" + explanation).

7. Running it

Frontend

```
cd tnpSC-mentor && npm install
# .env: VITE_SUPABASE_URL, VITE_SUPABASE_ANON_KEY
npm run dev      # local (http://localhost:5173)
npm run build
```

Data pipeline (`scrapers/`, needs `.env: SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, ANTHROPIC_API_KEY`)

```
pip install -r requirements.txt
python tnpSCmaster_scraper.py      # (or aptitude/pyq/samacheer/current_affairs/tnpscgu
python upload_new.py <file>.json  # dedup-insert (incl. *_ta)
python consolidate.py             # paraphrase -> why_wrong -> Tamil -> dedupe
python export_questions.py && python make_pdfs.py  # regenerate docs
```

Automation - `.github/workflows/monthly-ca.yml` runs `monthly_ca_update.py` on the 5th monthly (repo secrets `SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY`).

Roles - set `profiles.role = 'admin'` for view-only question-bank mode.

8. Copyright

Most content is scraped from third-party practice sites. Mitigations in place: explanations are **our own** (AI-written), and stems are **paraphrased** into

original wording; genuine past-paper items kept verbatim; `source_url` retained for provenance. See **HANDOVER-CLIENT.md** for the plain-English version and the pre-monetisation checklist.

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